

The Machine Learning Landscape

AI Bootcamp in Biotechnology – Rutgers and KAIMRC Academy

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Things you should know about this bootcamp

- ▶ Lots of diverse material and new concepts in a single day.
 - ▶ Programming and machine learning are **NOT** spectator sports – you need to practice these skills over and over again.
- ▶ All materials are on GitHub:
 - ▶ https://github.com/wevanjohnson/2026_09_Saudi_AI
- ▶ Each lecture has a matching hands-on workshop to work through on your own.
- ▶ Helpful background:
 - ▶ Introductory statistics and molecular biology
 - ▶ Basic R programming: tidyverse, ggplot2, R Markdown
- ▶ Examples are in R – the concepts are language-agnostic, and everything here has a scikit-learn equivalent.

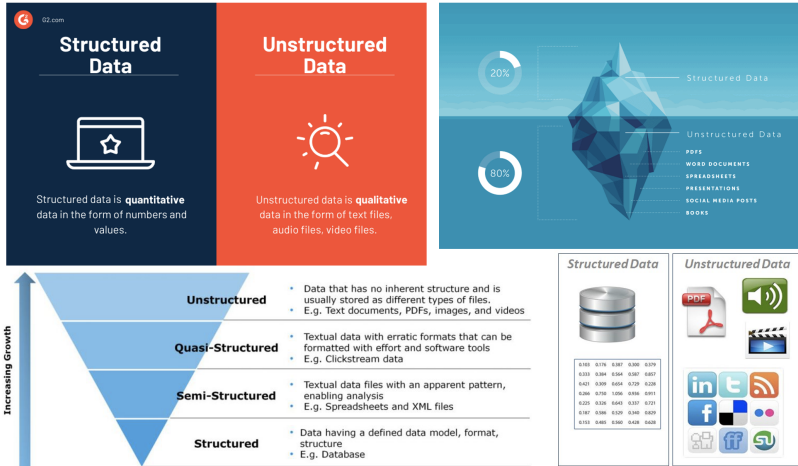
Introduction to Machine Learning and Data Science

BIG DATA

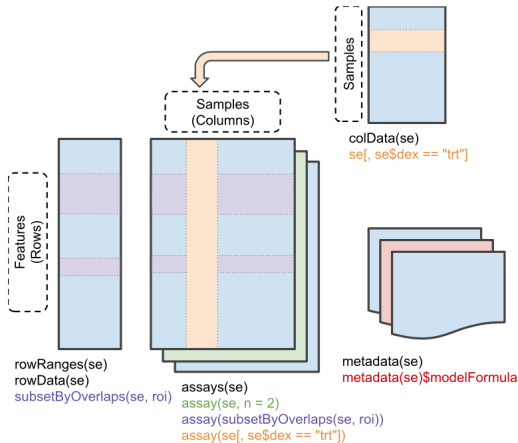


Big Data has fundamentally changed how we look at science and business. Along with advances in analytic methods, they are providing unparalleled insights into our physical world and society

Structured vs. Unstructured data

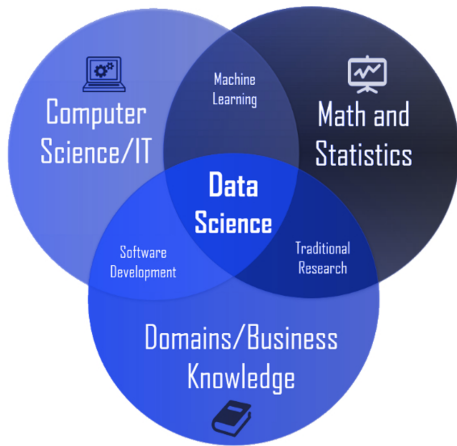


Structured vs. Unstructured data



Summarized Experiment

Data Science Revolution



- ▶ Few have all the skills
- ▶ Flexibility in area (business, strategy, health care) and conditions
- ▶ Data science makes companies and data better!

MODERN DATA SCIENTIST

Data Scientist, the sexiest job of 21th century requires a mixture of multidisciplinary skills ranging from an intersection of mathematics, statistics, computer science, communication and business. Finding a data scientist is hard. Finding people who understand who a data scientist is, is equally hard. So here is a little cheat sheet on who the modern data scientist really is.

MATH & STATISTICS

- ☆ Machine learning
- ☆ Statistical modeling
- ☆ Experiment design
- ☆ Bayesian inference
- ☆ Supervised learning: decision trees, random forests, logistic regression
- ☆ Unsupervised learning: clustering, dimensionality reduction
- ☆ Optimization: gradient descent and variants

DOMAIN KNOWLEDGE & SOFT SKILLS

- ☆ Passionate about the business
- ☆ Curious about data
- ☆ Influence without authority
- ☆ Hacker mindset
- ☆ Problem solver
- ☆ Strategic, proactive, creative, innovative and collaborative



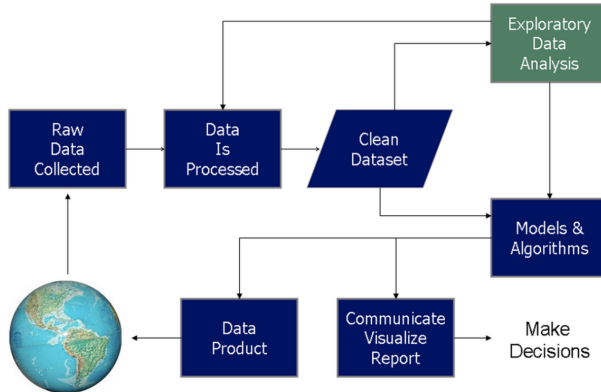
PROGRAMMING & DATABASE

- ☆ Computer science fundamentals
- ☆ Scripting language e.g. Python
- ☆ Statistical computing package e.g. R
- ☆ Databases SQL and NoSQL
- ☆ Relational algebra
- ☆ Parallel databases and parallel query processing
- ☆ MapReduce concepts
- ☆ Hadoop and Hive/Pig
- ☆ Custom reducers
- ☆ Experience with xaaS like AWS

COMMUNICATION & VISUALIZATION

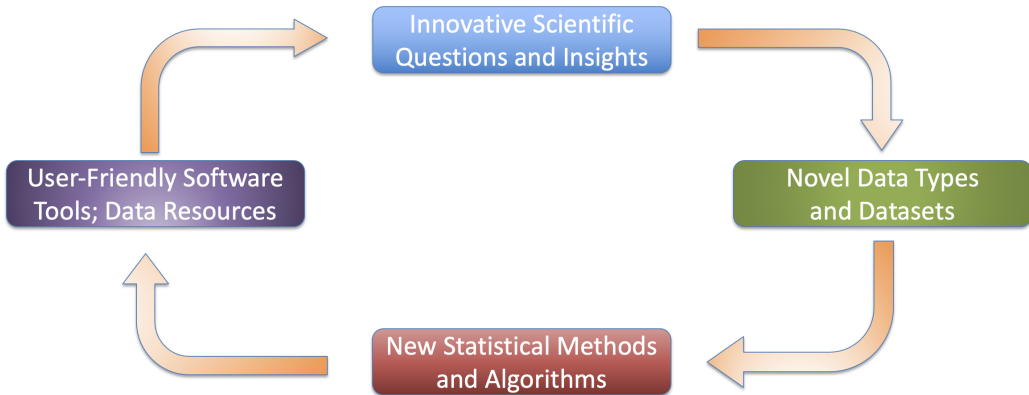
- ☆ Able to engage with senior management
- ☆ Story telling skills
- ☆ Translate data-driven insights into decisions and actions
- ☆ Visual art design
- ☆ R packages like ggplot or lattice
- ☆ Knowledge of any of visualization tools e.g. Flare, D3.js, Tableau

Data Science Process



Scientific Cycle for Data Science

Johnson Lab Approach to Science:

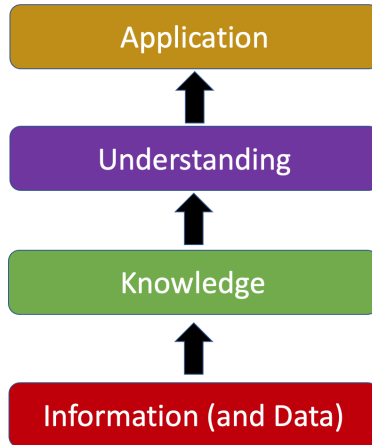


Keeping the “Science” in Data Science

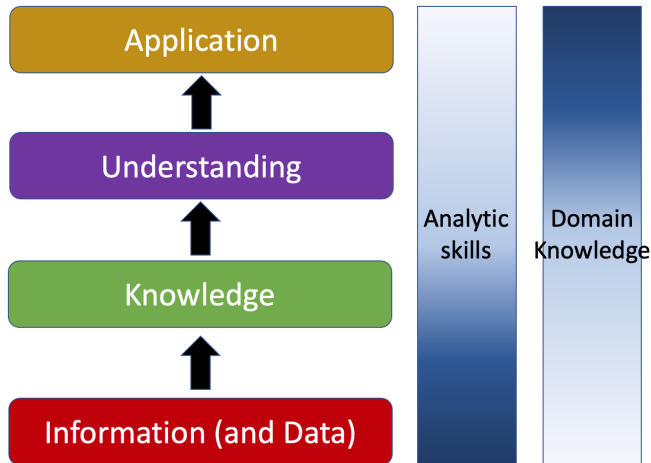
Domain Knowledge

Domain knowledge is knowledge of a specific, specialized discipline or field, in contrast to general (or domain-independent) knowledge. For example, in describing a software engineer may have general knowledge of computer programming as well as domain knowledge about developing programs for a particular industry. People with domain knowledge are often regarded as specialists or experts in their field.
(Wikipedia!)

Analytics Hierarchy



Analytics Hierarchy



Formal definitions

Machine learning is a computer's way of learning from examples, and it is one of the most useful tools we have for the construction of *artificially intelligent* systems

Artificial intelligence is a term used when machines (or computers) can mimic functions that humans can do.

- ▶ Example: Learning and problem solving

Formal definitions

An **Algorithm** is a sequence of actions that are “self-contained”

- ▶ Effective method for calculation
 - ▶ Finite steps/instructions
 - ▶ When applied, it produces a correct answer
 - ▶ The instructions need to be followed
 - ▶ In principle, it can be done by a “human”

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Formal definitions

- ▶ Computer Code:
 - ▶ Human readable text
 - ▶ Fully executable description of a software system
- ▶ What's in the data?
 - ▶ Datum: “(thing) given”
 - ▶ Data is useful only when it has been analyzed

Supervised vs unsupervised learning

Machine learning algorithms are generally classified into two categories:

1. **Supervised:** Outcomes used to create the predictor, e.g., classification.
2. **Unsupervised:** Don't know outcomes, rather interested in discovering groups, e.g., clustering.

Choosing a method for your research question

If you are asking. . .	Start with
Which group does this sample belong to?	Classification – logistic regression, SVM, random forest, XGBoost
How much, how many, how long?	Regression – linear and penalized regression, regression trees, GBM
Are there subgroups I did not define?	Clustering – k -means, hierarchical clustering
Can I see structure in high dimensions?	Dimension reduction – PCA, NMF, UMAP
Which variables actually matter?	Feature selection – LASSO, elastic net, variable importance, SHAP
Images, free text, or sequences?	Deep learning – CNNs, RNNs, transformers

What actually decides it

The question narrows the choice. Four constraints usually settle it:

1. **Is the outcome labeled?** No labels means unsupervised, whatever you had hoped to do.
2. **How many samples per feature?** When $p \gg n$, penalized and kernel methods will beat deep learning.
3. **Do you need to explain the model?** A clinician, a regulator, or a reviewer may need the reason, not just the prediction.
4. **How much signal do you expect?** Flexible models find structure whether or not it is there – which is what the evaluation session is about.

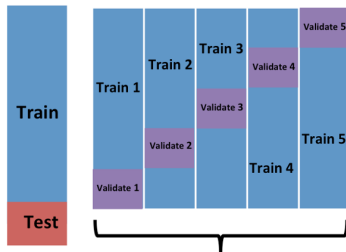
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Cross-validation: Iterative retraining of the predictor on multiple partitions of training set



Use cross validation to optimize model parameters

Formal definitions

Confusion matrix: tabulates each combination of predicted and actual values:

	Actually Positive	Actually Negative
Predicted positive	True positives (TP)	False positives (FP)
Predicted negative	False negatives (FN)	True negatives (TN)

Overfitting: Dangerously over-optimistic assessments—this is a *big problem* in machine learning

The caret package in R

The caret package in R has several useful functions for building and assessing machine learning methods.

▶ *Examples:*

Regularization in Machine Learning

In regression analysis, the features are estimated using coefficients while modeling. In small sample sizes or noisy data coefficient estimates could be anecdotally incorrect (e.g., overfitting) or innacurate.

If the estimates can be restricted, penalized, or shrunk towards zero, then the impact of insignificant features might be reduced and would prevent models from high variance with a stable fit.¹

¹Adapted from: <https://www.analyticssteps.com/blogs/l2-and-l1-regularization-machine-learning>

Regularization in Machine Learning

Regularization is the most used technique to penalize complex models in machine learning, it is deployed for reducing overfitting (or, contracting generalization errors) by putting small network weights into the model (adding a small amount of bias). Also, it enhances the performance of models for new inputs.²

²Source: <https://www.analyticssteps.com/blogs/l2-and-l1-regularization-machine-learning>

Regularization in Machine Learning

Examples of regularization in machine learning, include:

- ▶ Regression: Penalizing coefficients to create parsimonious models (variable selection)
- ▶ K-means: Restricting the segments for avoiding redundant groups.
- ▶ Neural networks: Confining the complexity (weights) of a model.
- ▶ Random forests: Reducing the depths of tree and branches (new features)

Hands-on practice

Everything in this session has a matching workshop you can work through at your own pace:

`workshop_1`

- ▶ Dimension reduction: PCA and UMAP
- ▶ Building up a plot with `ggplot2`
- ▶ Comparing differential expression methods

All workshops are in the course repository, with the R Markdown source and a rendered HTML version.

Session info

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.6.2
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/libBLAS.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.1
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Asia/Qatar
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices utils      datasets  methods   base
##
## loaded via a namespace (and not attached):
## [1] compiler_4.5.2    fastmap_1.2.0     cli_3.6.6         tools_4.5.2
## [5] htmltools_0.5.9   otel_0.2.0        rstudioapi_0.19.0 yaml_2.3.12
## [9] rmarkdown_2.31    knitr_1.51        xfun_0.58         digest_0.6.39
## [13] rlang_1.2.0       evaluate_1.0.5
```